

THE IRRATIONALITY OF THE 5-ADIC ZETA VALUE $\zeta_5(3)$

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ABSTRACT. We prove that the Kubota–Leopoldt 5-adic zeta value $\zeta_5(3) \in \mathbb{Q}_5$ is irrational, with the effective bound $|\zeta_5(3) - p/q|_5 > \max(|p|, |q|)^{-390}$ for $\max(|p|, |q|)$ large. The proof is the arithmetic holonomy method of Calegari, Dimitrov and Tang, in the form they used for $\zeta_2(5)$: Calegari’s p -adic Apéry construction on $X_0(5)$ gives an overconvergent function H of 5-adic radius 5^3 whose coefficients would be rational with denominators $[1, \dots, n]^3$ if $\zeta_5(3)$ were rational, and a holonomy bound with an archimedean template chosen among *all* holomorphic self-maps of the q -disc caps the dimension of the space such functions span below the four exhibited. The multi-place form of the bound (with p -adic radii) is derived from the published Bost–Charles form by a twist of the integral structure in Bost’s slopes inequality. Calegari’s original criterion fails for this value by 11%; the crude sup-norm holonomy bound fails by 1.55 nats; the refined bound succeeds by 0.074 nats.

1. STATEMENT

For a prime p , let ζ_p denote the Kubota–Leopoldt p -adic zeta function and, for $k \geq 1$, $\zeta_p(2k+1) \in \mathbb{Q}_p$ its value at the positive odd integer $2k+1$ (interpolating $\zeta^*(s) = (1-p^{-s})\zeta(s)$ at negative integers of the right parity). Calegari [5] proved $\zeta_p(3) \notin \mathbb{Q}$ for $p = 2, 3$ and $L_2(2, \chi_{-4}) \notin \mathbb{Q}$; Calegari–Dimitrov–Tang [7] proved $\zeta_2(5) \notin \mathbb{Q}$ (see also Lai–Sprang–Zudilin [12] for an Apéry-type proof); Beukers [1] proved $\zeta_p(2), \zeta_p(3) \notin \mathbb{Q}$ for $p = 2, 3$ and obtained, for $p = 5$, only that $\zeta_5(3) \pm L_5(3, \chi_5)$ is irrational; Lai–Lupu–Sprang [11] proved that for every $p \geq 5$ some $\zeta_p(i)$ with odd $i \in [3, p + p/\log p + 5]$ is irrational.

Theorem 1.1. $\zeta_5(3) \notin \mathbb{Q}$. Moreover, for all $p/q \in \mathbb{Q}$ with $\max(|p|, |q|)$ sufficiently large,

$$\left| \zeta_5(3) - \frac{p}{q} \right|_5 > \max(|p|, |q|)^{-390}.$$

The characterisation of $\zeta_5(3)$ used throughout is the one of [5, §2.2], [7, §6.2]: $\zeta_5(3)/2$ is the unique element $\eta \in \mathbb{Q}_5$ such that

$$(1) \quad E_{-2}^* := \eta + \sum_{n \geq 1} \left(\sum_{d|n, 5 \nmid d} d^{-3} \right) q^n$$

is the q -expansion of an overconvergent 5-adic modular form of weight -2 and level $\Gamma_0(5)$.

2. THE METHOD

2.1. Calegari’s construction. Let $x = (\Delta(5\tau)/\Delta(\tau))^{1/4} = q \prod_{n \geq 1} ((1 - q^{5n})/(1 - q^n))^6 \in q + q^2\mathbb{Z}[[q]]$, a Hauptmodul of $X_0(5)$ with $x = 0$ at the cusp ∞ and $x = \infty$ at the cusp 0 ; its formal inverse lies in $q \in x + x^2\mathbb{Z}[[x]]$. Let $E_2^* = E_2(\tau) - 5E_2(5\tau) \in 1 + q\mathbb{Z}[[q]]$ (here $4k/B_{2k} = 24$ makes E_2 integral), and set

$$H := E_2^* \cdot E_{-2}^* = \sum_{n \geq 0} h_n x^n, \quad h_n = a_n + \eta b_n, \quad a_n, b_n \in \mathbb{Q}.$$

H is a weight-0 overconvergent form, i.e. a function on the rigid curve $X_0(5)$. The Fricke involution sends $x \mapsto 5^{-3}/x$ and exchanges the two components $\{|x|_5 \leq 1\}$, $\{|x|_5 \geq 5^3\}$ of the ordinary locus; by Buzzard's theorem [4, Thm. 5.2] the slope-0 form E_{-2}^* extends over the whole supersingular annulus, so $H(x)$ converges on $|x|_5 < 5^3$. (We also read this off the coefficients: $v_5(h_n) - 3n$ is $-O(\log n)$ and never positive for $n \leq 2000$; see §4.)

Suppose $\zeta_5(3) \in \mathbb{Q}$. Then $H \in \mathbb{Q}[[x]]$.

2.2. The holonomy bound. We use the notation of [6, Thm. 2.5.1]: m functions $f_1, \dots, f_m \in \mathbb{Q}[[x]]$, $\mathbb{Q}(x)$ -linearly independent, with denominator types encoded by a step-shaped array $\mathbf{b} = (b_{i,j})$ (row sums σ_i , u_j the number of zero entries in column j) and

$$\tau(\mathbf{b}) = \frac{1}{m^2} \sum_{i=1}^m (2i-1)\sigma_i;$$

a holomorphic $\varphi : (\mathbb{D}, 0) \rightarrow (\mathbb{C}, 0)$ such that each $f_i \circ \varphi$ is meromorphic on $\mathbb{D}$; and the Bost–Charles integral $\text{BC}(\varphi) = \iint_{\mathbb{T}^2} \log |\varphi(z) - \varphi(w)| d\mu(z) d\mu(w)$. Their theorem: if $\log |\varphi'(0)| > \tau(\mathbf{b})$ and the f_i are holonomic, then $m \leq \text{BC}(\varphi) / (\log |\varphi'(0)| - \tau(\mathbf{b}))$.

For a p -adic target one needs the version with p -adic radii, used in [7, §6.2] and attributed there to a forthcoming paper:

Theorem 2.1 (multi-place bound). *In the situation above, let S be a finite set of primes and for $p \in S$ let $R_p = p^{\alpha_p}$, $\alpha_p \in \mathbb{Z}_{>0}$, be such that the coefficients $c_{i,n}$ of x^n in f_i satisfy $|c_{i,n}|_p \leq Cn^A R_p^{-n}$ for all i, n . If $\log |\varphi'(0)| + (1 - \frac{1}{m}) \sum_{p \in S} \log R_p > \tau(\mathbf{b})$ then*

$$m \leq \frac{\text{BC}(\varphi) + \sum_{p \in S} \log R_p}{\log |\varphi'(0)| + \sum_{p \in S} \log R_p - \tau(\mathbf{b})}.$$

A proof from published ingredients is given in §3. The crude (sup-norm) analogue, with $\sup_{\mathbb{T}} \log |\varphi|$ in place of BC , is Dimitrov's arithmetic holonomicity theorem [8, 9]; it does not suffice here (Remark 6.2).

2.3. The inventory and the reduction to one inequality. By Kontsevich–Zagier [10, §2.3], a modular form of weight w expressed in a modular function satisfies a linear ODE of minimal order $w+1$; for the product H this gives an *inhomogeneous* ODE over $\mathbb{Q}(x)$ of minimal order 3, so that $\{1, H, \delta H, \delta^2 H\}$, $\delta = x d/dx$, are $m = 4$ $\mathbb{Q}(x)$ -linearly independent functions (Lemma 4.2). Their denominators are of type $[1, \dots, n]^3$ (Proposition 4.1), so $\mathbf{b}$ has $r = 3$, $b_{i,j} = 1$ for $i \geq 2$, $u_j = 1$, and $\tau(\mathbf{b}) = 3(1 - \frac{1}{16}) = \frac{45}{16}$. With $S = \{5\}$, $L := \log R_5 = 3 \log 5$, and $\rho := |\varphi'(0)|$, Theorem 2.1 contradicts $m = 4$ exactly when

$$(2) \quad \text{cost}(\varphi) := \text{BC}(\varphi) - 4 \log \rho < 3(L - 4 + \frac{1}{4}) = 3.2349412 \dots =: \text{budget}.$$

The budget is arithmetic ($p = 5$, $k = 1$); the cost depends only on the template.

2.4. Admissible templates. $H(x(q))$ and its δ -derivatives are meromorphic on the whole disc $|q| < 1$ (E_2^* is holomorphic on $\mathfrak{H}$, the coefficients of E_{-2}^* are bounded, and d/dx only introduces poles at the critical points of x). Hence $\varphi = x \circ \psi$ is admissible for *every* holomorphic $\psi : \mathbb{D} \rightarrow \mathbb{D}$ with $\psi(0) = 0$ — no injectivity is required and $\psi(\mathbb{D})$ need not be a Jordan domain. Writing $\psi(z) = z \exp(c_0 + \sum_{k \geq 1} c_k z^k)$ with real c_k , the only constraint is $\max_{|z|=1} \log |\psi| \leq 0$, and $\log \rho = c_0$ exactly (since $x'(0) = 1$). We optimise (2) over this family. The points of $|q| = 1$ at which x is unbounded are the $\Gamma_0(5)$ -images of the cusp 0, which are dense on the circle (unlike $p = 2$, where CDT's contour may touch $q = -1$); the optimum stays inside $|q| \leq 0.8363$ with the constraint inactive.

3. PROOF OF THEOREM 2.1

We follow the proof of [6, Thm. *main:BC form*, §6.3], which runs Bost's slopes inequality [2] on the filtered evaluation module of auxiliary polynomials, and change only the integral structure at the primes of S . Recall their setup: E_D is the lattice of $\sum_{i=1}^m Q_i f_i$ with $Q_i \in \mathbb{Z}[1/x]_{\leq D}$ (weighted by the lcm factors prescribed by $\mathbf{b}$), filtered by vanishing order n , with the evaluation maps $\psi_D^{(n)} : E_D^{(n)} / E_D^{(n+1)} \rightarrow G^{(n)} = \mathbb{Z} \cdot x^{n-D}$ onto the leading coefficient; the archimedean norm on E_D is the Bost–Charles metric attached to φ , and $\|x^{n-D}\|_\infty = 1$. Bost's inequality reads

$$(3) \quad \widehat{\deg} \overline{E_D} \leq \sum_{n \in \mathcal{V}_D} \left[\widehat{\mu}_{\max}(\overline{G^{(n)}}) + h_\infty(\psi_D^{(n)}) + h_{\text{fin}}(\psi_D^{(n)}) \right],$$

where $\mathcal{V}_D$ is the set of attained vanishing orders, $\#\mathcal{V}_D = m(D+1)$ and $\sum_{n \in \mathcal{V}_D} n = \frac{m^2}{2} D^2 + o(D^2)$ by their Kolchin-type lemma. CDT evaluate every term: $\widehat{\deg} \overline{E_D} = (\frac{m}{2} \text{BC}(\varphi) + \dots) D^2 + o(D^2)$, $\sum h_\infty \leq (-\frac{m^2}{2} \log |\varphi'(0)| + m \text{BC}(\varphi)) D^2 + o(D^2)$, and $\sum h_{\text{fin}}$ accounts for the lcm denominators and produces $\tau(\mathbf{b})$.

The twist. For $p \in S$ replace the two lattices by

$$\Lambda_D := \bigoplus_{j=0}^D \mathbb{Z}(R_p x)^{-j} \supset \mathbb{Z}[1/x]_{\leq D}, \quad G_{\text{ad}}^{(n)} := \left(\prod_{p \in S} R_p^{n-D} \right) \mathbb{Z} \cdot x^{n-D},$$

i.e. use Λ_D in place of $\mathbb{Z}[1/x]_{\leq D}$ in each of the m summands of E_D , keeping every archimedean norm unchanged. Since $E_{D,\mathbb{R}}$ and $\text{Gr}_n(F)_{\mathbb{R}}$ and their Euclidean norms are unchanged and h_v depends only on the v -adic norms, all archimedean estimates of [6] hold verbatim; at primes $\ell \notin S$ nothing changes (R_p is an ℓ -unit). Three computations account for the twist.

- (1) $\Lambda_D \supset \mathbb{Z}[1/x]_{\leq D}$ has index $\prod_{p \in S} R_p^{D(D+1)/2}$, so $\widehat{\deg} \overline{E_D^{\text{ad}}} = \widehat{\deg} \overline{E_D} + \frac{m}{2} D^2 \sum_p \log R_p + o(D^2)$.
- (2) $\overline{G_{\text{ad}}^{(n)}}$ has rank one with generator of archimedean norm $\prod_p R_p^{n-D}$, so $\widehat{\mu}_{\max}(\overline{G_{\text{ad}}^{(n)}}) = -(n-D) \sum_p \log R_p$, and summing over $\mathcal{V}_D$ gives $-(\frac{m^2}{2} - m) D^2 \sum_p \log R_p + o(D^2)$.
- (3)

Lemma 3.1 (p -adic operator norm). *For $p \in S$ and every n , $h_p(\psi_D^{(n)}) \leq \log(Cn^A)$.*

Proof. The p -adic norm on E_D^{ad} is $\|\sum_{i,j} q_{ij} f_i x^{-j}\|_{E,p} = \max_{i,j} |q_{ij}|_p R_p^{-j}$ (the p -parts of the lcm factors are $p^{O(\log D)}$ and are absorbed into the $o(D^2)$ of CDT's finite-place estimate), and on $G_{\text{ad}}^{(n)}$ it is $\|c x^{n-D}\|_{G,p} = |c|_p R_p^{n-D}$. The image of $Q = \sum q_{ij} f_i x^{-j}$ is $c_n(Q) x^{n-D}$ with $c_n(Q) = \sum_{i,j} q_{ij} c_{i,n-D+j}$, so by the ultrametric inequality and the hypothesis on the $c_{i,n}$,

$$|c_n(Q)|_p \leq \max_{i,j} |q_{ij}|_p C n^A R_p^{-(n-D+j)} = C n^A R_p^{-(n-D)} \|Q\|_{E,p},$$

whence $\|\psi_D^{(n)}[Q]\|_{G,p} \leq C n^A \|Q\|_{E,p}$; as $c_n(Q)$ depends only on the class of Q , the same holds for the quotient norm. $\square$

Summed over $\mathcal{V}_D$, (3) is $O(D \log D)$. Inserting (1)–(3) into (3) and dividing by D^2 adds $\frac{m}{2} \Lambda$ to the left and $-(\frac{m^2}{2} - m) \Lambda$ to the right of CDT's inequality, where $\Lambda = \sum_p \log R_p$; after their optimisation over the auxiliary parameters this is $m(\log |\varphi'(0)| - \tau(\mathbf{b})) + (m-1) \Lambda \leq \text{BC}(\varphi)$, which is the statement. $\square$

Remark 3.2. With $m = 2$, $\mathbf{b} = 0$ and φ univalent, Theorem 2.1 reduces to the Borel–Dwork criterion $\rho \prod_p R_p > 1$; the weight $1 - \frac{1}{m}$ on Λ is forced by the product formula (the margin is invariant under $x \mapsto x/p^v$). The high-dimensional proof of the rearrangement-numerator form

[6, Thm. *elementary form*] can be twisted in the same way but, as far as we can see, only yields the weight $(1 - \frac{1}{m})^2$; we therefore use the Bost–Charles numerator throughout.

4. THE ARITHMETIC INPUTS

Proposition 4.1 (denominator type). *Write $H = \sum h_n x^n$ with $\zeta_5(3) \in \mathbb{Q}$ assumed, and let D'_n be the least common multiple of the integers $\leq n$ prime to 5. Then $(D'_n)^3 h_n \in \mathbb{Z}[1/5]$ after clearing a fixed denominator, and the exponent 3 is sharp. The same holds for $\delta H, \delta^2 H$ (the coefficients are $n^j h_n$).*

Proof. $E_2^* \in 1 + q\mathbb{Z}[[q]]$; the q -coefficients of E_{-2}^* are $\sum_{d|n, 5 \nmid d} d^{-3}$, whose denominators divide $(D'_n)^3$; and $q \in x + x^2\mathbb{Z}[[x]]$. Hence $(D'_n)^3 \cdot [x^n](E_2^* E_{-2}^*) \in \mathbb{Z}[1/5]$. Sharpness was verified exactly for $n \leq 60$. The use of δ rather than d/dx keeps $\delta^j H$ literally within the shape required by [6], while spanning the same $\mathbb{Q}(x)$ -module as $\{H, H', H''\}$. $\square$

The prime-to-5 lcm grows like e^n (prime number theorem), so the array is as stated in §2.3; the 5-power content of h_n is what R_5 measures.

Lemma 4.2 (independence). *$1, H, \delta H, \delta^2 H$ are $\mathbb{Q}(x)$ -linearly independent.*

Proof. A relation $a_0 + a_1 H + a_2 \delta H + a_3 \delta^2 H = 0$ with $a_i \in \mathbb{Q}(x)$ not all zero and $(a_1, a_2, a_3) \neq 0$ is an inhomogeneous linear ODE for H of order ≤ 2 , contradicting the minimal order 3 of Kontsevich–Zagier (structurally: $\theta^3 E'_{-2} = E_4 - E_4(5\tau)$, the Eichler cocycle has degree ≤ 2 , and E_2^* is a period of the Sym^2 of the Picard–Fuchs system); $(a_1, a_2, a_3) = 0$ forces $a_0 = 0$. The rank was also verified directly: no relation with polynomial coefficients of degree ≤ 120 exists (computed modulo $2^{61} - 1$). $\square$

Proposition 4.3 (5-adic radius). $R_5 = 5^3$.

Proof. Buzzard [4, Thm. 5.2] (slope 0, Eisenstein family) gives analyticity of E_{-2}^* on the ordinary component containing ∞ together with the entire supersingular annulus, which in the coordinate x is the disc $|x|_5 < 5^3$; cf. [5, §3] for $p = 2, 3$ and [7, §6.2]. Independently, exact computation of h_n in $\mathbb{Z}/5^{6150}$ for $n \leq 2000$ (cross-checked against exact rationals for $n \leq 600$) gives $v_5(h_n) \geq 3n - 0.974 - 1.728 \log n$ with $\max_n (v_5(h_n) - 3n) = -3$, so the radius is exactly 5^3 . (Each factor $E_2^*(x)$, $E_{-2}^*(x)$ alone has radius 1; the overconvergence belongs to the product, which is a function on the curve.) $\square$

5. THE TEMPLATE AND THE NUMERICAL CERTIFICATE

Take $\psi(z) = z \exp(\sum_{k=0}^{40} c_k z^k)$ with

$$\begin{aligned} c_{0..5} &= (-0.531289158, -0.225379074, -0.127598877, -0.001824369, -0.027735780, -0.041418343) \\ c_{6..11} &= (+0.051444137, +0.026087140, -0.010853938, +0.001314954, -0.003547849, -0.023294126) \\ c_{12..17} &= (+0.000125751, +0.020158067, -0.001454440, -0.004262361, +0.007019143, -0.000198726) \\ c_{18..23} &= (-0.007684408, -0.006339902, +0.005397468, +0.001260337, +0.000134428, +0.014051878) \\ c_{24..29} &= (-0.011391873, -0.005386029, -0.000202917, +0.001464054, -0.000123972, +0.008670335) \\ c_{30..35} &= (-0.000051833, -0.006877094, -0.001450246, -0.001060183, +0.002930635, +0.002103211) \\ c_{36..40} &= (+0.001750273, -0.002516914, -0.002244428, +0.000524421, +0.001281738). \end{aligned}$$

Admissibility. On $|z| = 1$, $\log |\psi| = u(t) = c_0 + \sum c_k \cos 2\pi kt$ has Lipschitz constant $2\pi \sum k|c_k| = 2\pi \cdot 4.3589$; sampling u on 2^{22} points and adding the Lipschitz correction gives $\max u \leq -0.1788 < 0$, so $\psi(\mathbb{D}) \subset \mathbb{D}$ by the maximum principle. *Conformal size.* $\log \rho = c_0 = -0.531289158$ exactly. *The integrals.* With $g(t) = \log |x(\psi(e^{2\pi it}))|$, the a-priori bound $|g'| \leq 2\pi \max |\Lambda|(1 + \sum k|c_k|)$,

$\Lambda = q d \log x / dq = -\frac{1}{4}(E_2(q) - 5E_2(q^5))$, and the triangle-inequality majorant of the Lambert series on $|q| \leq 0.836266$, give $\text{Lip}(g) \leq 11427$; the rearrangement integral $\text{RE} = \int_0^1 2t g^*(t) dt$ on $N = 2^{24}$ nodes is then certified to $\pm \text{Lip}/N = \pm 6.8 \cdot 10^{-4}$. The Bost–Charles integral is computed on the same grid and is N -stable to 10^{-8} from 2^{12} to 2^{24} nodes (its convergence is $O(1/N)$ since φ is not injective on the boundary; we have no interval-arithmetic bound for it, and record both).

quantity	value	provenance
m	4	Lemma 4.2
$\tau(\mathbf{b})$	$45/16 = 2.8125$	Prop. 4.1
$\log R_5$	$3 \log 5 = 4.828313737$	Buzzard; measured to $n = 2000$
$\log \rho = c_0$	-0.531289158	exact
$\max q $ on contour	0.836265	
$\text{RE}(\varphi)$	1.066965 ± 0.00068	$N = 2^{24}$, a-priori error
$\text{BC}(\varphi)$	1.0355293	N -converged to 10^{-8}
bound (RE)	≤ 3.9716148	< 4 , margin $\geq +0.0421$
bound (BC)	3.9499804	< 4 , margin $+0.0743$
crude sup-norm form	5.714	fails (Remark 6.2)

Proof of Theorem 1.1. Assume $\zeta_5(3) \in \mathbb{Q}$. By §2 and §4, $\{1, H, \delta H, \delta^2 H\}$ satisfy the hypotheses of Theorem 2.1 with $S = \{5\}$, $R_5 = 5^3$, $\mathbf{b}$ as in §2.3 and $\varphi = x \circ \psi$; the entry condition $c_0 + \frac{3}{4} \cdot 3 \log 5 = 3.090 > \frac{45}{16}$ holds. Theorem 2.1 gives $4 \leq (\text{BC} + 3 \log 5) / (c_0 + 3 \log 5 - \frac{45}{16}) = 3.94998 < 4$, a contradiction. The measure follows from the quantitative form of the bound [7, §2.2–2.3] (as in their §6.2), with $\gamma = 1/4$ (one of the four functions has integer coefficients), $\rho_5 = 5^{-3}$ and the margin $\Delta_0 = 0.0742553$: $\kappa = \frac{3}{4}(mL + \sqrt{mL(mL - \Delta_0)}) / \Delta_0 = 389.8$ with $L = 3 \log 5$, $m = 4$. $\square$

6. REMARKS

Remark 6.1 (Calegari’s criterion). Calegari’s p -adic Apéry argument [5] compares the p -adic smallness of $b_n \eta - a_n$ with the archimedean size of the cleared coefficients; in his normalisation the criterion is $\theta_p > 1$ with $\theta_p = \sigma_p \log p / (k + \log \lambda_1)$, and for $(p, k) = (5, 1)$ one finds $\theta_5 = 0.8918$, his own recorded failure. The holonomy bound replaces the naive archimedean disc of radius $1/\lambda_1 = 5^{-3/2}$ by the template φ , whose conformal size $e^{c_0} = 0.588$ is 6.6 times larger, at the price of the numerator $\text{BC}(\varphi)$.

Remark 6.2 (the crude form fails). Dimitrov’s sup-norm bound $m \leq (\sup_{\mathbb{T}} \log |\varphi| + \sum \log R_p) / (\log |\varphi'(0)| + \sum \log R_p - \tau)$, optimised over discs and lune-bitten discs in the q -disc, gives at best $m \leq 5.71$ (cost 4.035 against budget 2.485), so it does not prove the theorem; the Bost–Charles numerator is essential. By contrast for CDT’s $\zeta_2(5)$ the crude form already suffices ($5.56 < 6$ in [9]).

Remark 6.3 (sensitivity). The verdict survives a 4% error in the numerator, $R_5 \geq 5^{2.991}$, or 0.35% in τ , but not a change of the weight $1 - \frac{1}{m}$ in Theorem 2.1 to $(1 - \frac{1}{m})^2$ (which would give margin -3.6); the weight is the one load-bearing step that is ours, and §3 is written so that it can be checked against [6, §6.3] line by line.

Remark 6.4 (other values). The same computation gives, for (p, k) with $p - 1 \mid 24$: $\zeta_7(3)$ fails by 2.05 nats, $\zeta_3(5)$ by 2.37, $\zeta_2(7)$ by 6.46, and $\zeta_5(5)$ fails the entry condition for every template ($3 \log 5 \cdot \frac{5}{6} < \frac{35}{36} \cdot 5$). It reproduces the known cases: Calegari’s $\zeta_2(3), \zeta_3(3), L_2(2, \chi_{-4})$ with large margins, CDT’s $\zeta_2(5)$ with their printed numbers (4.43206 and 4.48866 for their two contours, improved to 4.4389 here), and $L_3(2, \chi_{-3})$ on $X_0(3)$ (Beukers).

Remark 6.5 (novelty). We have not found $\zeta_5(3) \notin \mathbb{Q}$ in the literature cited above; in particular [1] reaches only $\zeta_5(3) \pm L_5(3, \chi_5)$ and [11] only the existence of an irrational value in a range. This draft is circulated for that check.

Reproducibility. All scripts and logs are in `lattice/padic_holonomy/` of the project repository; the full ledger of what is exact, numerically converged, and assumed is `consolidation/PADIC_HOLONOMY_CEN` and the proof of Theorem 2.1 with its checks is `consolidation/ADELIC_HOLONOMY.md` §2.6.

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